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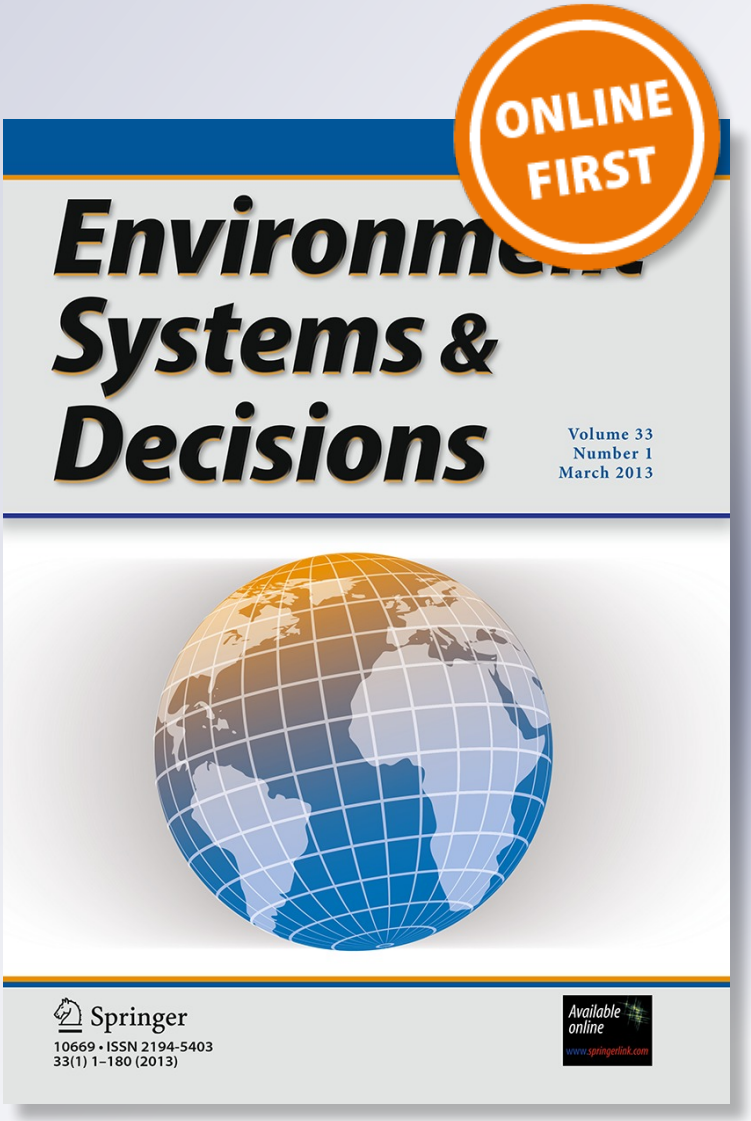

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Abstract This study uses Danxiashan, a world heritage site in Guangdong province, China, as a case study for planning a hypothetical geotourism network of heritage sites. This landscape has a multiplicity of values—its geoheritage cannot be separated from its ecological or cultural heritage. When designing a network of heritage sites for such a diversely valued landscape trade-offs must be made between differing and potentially conflicting objectives such as geotourism and geoconservation. To solve this multi-criteria decision problem, sites with potential value for people were designated as heritage sites, adapting the concept of geomorphosites—i.e., geomorphological heritage sites—to represent the intersection of anthropic values. In a GIS-based spatial decision support system heritage values for each site were weighted and ranked using the analytic hierarchy process and scenarios for alternative trail systems, and networks of tourism sites were generated by multi-criteria map overlay analysis and networking algorithms. The scenarios generated show how trade-offs can be made between oft-conflicting tourism and conservation objectives, how the design of parks can be optimized for multiple objectives, and how alternative design strategies can be explored. Such an approach could be used in scenario planning workshops to engage stakeholders in participatory design and consensus-based decision-making driven by geospatial science.

keywords Danxiashan · Tourism · Heritage conservation · Geomorphosites · Biogeomorphology · Spatial decision support systems · Fuzzy logic · Multi-criteria decision analysis

This research was presented as a lightning talk at the 2013 Geodesign Summit. The geospatial modeling and analysis used Esri's ArcGIS Desktop 10 with the Spatial Analyst, 3D Analyst, and Network Analyst extensions.

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1 Geomorphosites: prioritizing geoconservation

Landforms, scientifically, are valued foremost for their geological and geomorphological importance. A non-living analog to biodiversity—the variety of genes, species, and ecosystems (Wilson 2001)—geodiversity is the variety of geological, geomorphological, and soil forms and processes (Gray 2008). This diversity is valued for its aesthetics, natural and cultural heritage, economic value, and scientific importance (Gray 2008). The human values attributed to geodiversity are collectively referred to as geoheritage. Quantifying geodiversity or its value—its geoheritage—is a theoretical problem whereas prioritizing and optimizing the conservation of geoheritage, that is, geoconservation, is an applied problem. To practically conserve our geoheritage, the relative importance of these values must be identified in geographic places. A geomorphosite—a geomorphological heritage site—is "a landform to which a value can be attributed" (Panizza 2001). Geomorphosites should be identified, evaluated, and ranked so that conservation efforts can be prioritized. Methods and metrics for systematically quantifying and

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qualifying geoheritage are being developed so that geomorphosites can be evaluated and conserved (Panizza 2001; Reynard and Panizza 2005; Reynard et al. 2007; Ruban 2010).

Geomorphosites include landscapes like Danxiashan in Guangdong Province, China, geomorphological systems like the Mammoth–Flint Ridge Cave System in Kentucky, USA, or individual landforms like the Giant's Causeway in Country Antrim, UK. Just as a landscape can contain landforms, geomorphosites can contain smaller, nested geomorphosites. In Danxiashan there is a myriad of landforms including peak forests, cuestas, mesas, hoodoo columns, joints, caves, ravines, canyons, and valleys; each is a geomorphosite. Each site may be valued for different reasons. In Danxiashan a vertical joint with exposed strata may be valued primarily as a scientific record of the landscape's geomorphological evolution, a cave may be valued primarily for its history as a refuge for Taoist hermits; geocultural places like Yangyuanshi, a hoodoo column, may be valued primarily for their symbolism.

Geosites and geomorphosites can be assessed through either purely scientific, geophysical values or though a broad range of scientific and socio-cultural values. Ruban has proposed a purely scientific method for systematically quantifying the geoheritage of geosites (Ruban 2010). Just as metrics like richness, rarity, and endemism are used to quantify biodiversity and identify biodiversity hotspots (Myers et al. 2000), geosites can be assessed with metrics such as geodiversity, georichness, and geoabundance (Ruban 2010). In a critique of this method, Knight argues for a multidimensional evaluation of geomorphosites based on both quantitative and qualitative assessments of a wide range of scientific and socio-cultural values (Knight 2010).

Metrics for geodiversity alone will not represent all of the ways that geoheritage is valued. In addition to geological characteristics like geomorphology, stratigraphy, paleontology, paleogeography, sedimentology, and hydrology, the scientific value of a site may also depend upon its potential for research or education. Furthermore, geodiversity plays a fundamental role in structuring ecosystem and biospheric complexity (Gray 2008; Stallins 2006). The complex interactions that arise from geodiversity shape ecosystems and biogeomorphological systems, forming peculiar process-form interactions ranging from the terrestrial island biogeography of Danxiashan (Wu et al. 2008) to the formation of Chenier Plains in the Mississippi Delta (McBride et al. 2007). In a biogeomorphological system ecological patterns and processes interact with geomorphological process and forms (Viles 2011). If there are bio-geomorphic interactions at a site then ecological parameters should be considered alongside geological parameters when assessing its scientific value.

Socio-cultural values may be attributed to a landform or a landscape; a multiplicity of values ranging from culture and aesthetics to economics may play formative roles in shaping its geoheritage. A marriage of culture and landscape transforms space into place, imbuing a geographic space with cultural meaning and human value (Basso 1996). Furthermore, the context in which a geomorphosite is situated may also influence its anthropic valuation. Thus, the greater landscape within which a geomorphosite is nested may imbue the site with further socio-cultural meaning. A landform can be valued for its aesthetics—the strange, sculptural grandeur of a Danxia peak, for example—and it can also be valued for the aesthetics of its setting, for its views—views, for example, of a meandering river and verdant subtropical monsoon forest in the foreground, of mist-veiled peak forests in the distance.

Because of their socio-cultural and economic value geomorphosites are often exploited for mineral extraction or tourism. Extraction can be ruinous to a site and may degrade or destroy its scientific values. Tourism can also degrade the integrity of a site. The pulse of tourists during peak times can increase erosion and soil compaction (Shi 2006), accelerate the wearing of historic structures, heighten noise pollution and edge effects (Forman 1995), and threaten water quality (Zhong et al. 2008). Overcrowding can degrade visitors' aesthetic and cultural experience. As theorized in the tourism area lifecycle model when overheated tourism exceeds the carrying capacity of the landscape the integrity and viability of the landscape can be severely threatened (Butler 2005; Zhong et al. 2008). If a heritage landscape is overexploited and overdeveloped, then air and water quality may deteriorate, the landscape may become fragmented, landforms destroyed or compromised, habitats lost, and viewsheds degraded (Zhong et al. 2008). Tourism, however, can also spread awareness of the site's values, educate the visiting public, and build public support and political will for geoconservation. Thus, heritage conservation and tourism may have conflicting objectives, but the potential for synergy.

Panizza advocates a multidimensional approach for evaluating geomorphosites. He argues that a site should be evaluated by scientific, socio-economic, scenic, and cultural criteria and details a methodology for quantifying a site's scientific value based on geomorphological, paleogeomorphological, ecological, and educational sub-criteria (Panizza 2001). While also supporting a multidimensional approach, Reynard argues that geomorphological values should be considered separately from other, additional values. He proposes that core geomorphological values—rarity, representativeness, paleogeography, and integrity—should be objectively quantified. Additional values—ecology, culture, aesthetics, and economics—should then assessed to show the links between geomorphological values and their context, but no composite is calculated—so that the distinction between them is maintained

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(Reynard et al. 2007). However, a focus on scientific metrics that overlooks socio-cultural values may lead to ineffectual geoconservation, failing to muster public support or build political will. Pena dos Reis and Henriques argue that geoheritage evaluations should avoid either a scientific or social bias. Whereas a socially biased evaluation, prejudiced toward iconographic and symbolic values, may underrepresent the international relevance of the site, a scientifically biased evaluation, prejudiced toward documental and conceptual values, is less likely to generate public interest and support. Public understanding and interest are requisite for the building the political support necessary for practical geoconservation (Pena dos Reis and Henriques 2009).

Geomorphosite assessments for geoconservation and geotourism planning should be based upon multiple criteria, account for scale and nestedness (Knight 2010), and be custom tailored to their context (Reynard et al. 2007). Conservation and tourism objectives are context dependent; multi-criteria geomorphosite assessments should be defined by local socio-cultural perspectives and international scientific knowledge. Therefore, geomorphosite assessment and planning requires context-specific, bespoke strategies directed by international scientific guidance.

Quantifying geoheritage is a multi-criteria decision problem. To prioritize conservation, factors ranging from geology, geomorphology, ecology, biogeomorphology, economics, culture, and aesthetics may need to be considered, depending upon the context. Depending upon the objective, these criteria may be valued and thus weighted differently.

Appropriately weighting multiple criteria can be great conceptual challenge—large numbers of criteria or radically different criteria are difficult to rationally compare and multiple stakeholders may value criteria differently. Multi-criteria decision analysis techniques like the analytic hierarchy process can be used to hierarchically structure decision problems, compare preferences, make trade-offs, weigh criteria, and assess inconsistency in geoconservation scenarios. In the analytic hierarchy process a reciprocal matrix of pairwise comparisons is used to determine the weights of criteria by calculating the eigenvector, that is, the vector of preference, for a given set of alternatives (Saaty 1980; Pedrycz and Song 2011). Geomorphosite assessments based on multi-criteria decision analysis can be used to rationally compare multiple criteria based upon context-specific objectives and prioritize geoconservation and geotourism efforts.

2 Beyond geomorphosites: aims and objectives

A recent line of research—geosites and geomorphosites—seeks to quantify geodiversity and prioritize geoconservation. Building on this theoretical research, we developed an applied method for spatially prioritizing and optimizing conservation planning for heritage landscapes. Our method moves beyond past research about geomorphosites by evaluating and prioritizing both scientific and sociocultural criteria. Furthermore we have demonstrated how prioritization can be used to spatially optimize the structure of a heritage landscape.

The objectives of this research were to create a process to:

1. comparably quantify multiple anthropic values in a landscape,
2. prioritize conservation and tourism in a landscape based on the objectives of multiple stakeholders and decision-makers,
3. and design an optimized trail network that maximizes scientific, educational, and socio-cultural opportunities while minimizing impacts on the landscape.

This process was tested with a series of sustainable tourism scenarios for Danxiashan, a UNESCO World Heritage site with notable bio-geomorphic interactions (Wu et al. 2008; Peng et al. 2008; Hou et al. 2008). Despite already having been developed for tourism, Danxiashan was selected for this case study so that the process could be tested with a real-world site with well-documented heritage values. While Danxiashan already has extensive tourist infrastructure, several scenarios for alternative designs were explored. Such scenarios could contribute to the future planning and management of this World Heritage site.

3 Danxiashan: a heritage landscape

Danxiashan (Fig. 1), our geoconservation case study, was listed in 2010 as part of a serial World Heritage site, the China Danxia, on account of its aesthetics and as a record of the earth's history, World Heritage criteria vii and viii. Just one of the six properties in the 82,151 ha China Danxia listing, Danxiashan itself covers 16,800 ha and has a 12,400 ha buffer area (Ministry of Housing and Urban-Rural Development of the People's Republic of China 2008; UNESCO 2010). Danxiashan—a red-bed landscape dominated by peak forests and ravines that are structured by crossing NNE–EW fault blocks—has a rich legacy of geodiversity, biodiversity, and cultural heritage (Ministry of Housing and Urban-Rural Development of the People's Republic of China 2008; Peng 2001). Danxiashan is a biogeomorphic system; it has been shaped by coupled biological and geomorphological patterns and processes. Biogeomorphology is the interrelationship between ecology and geomorphology; it is an important determinant of landscape form and function (Goudie and Viles 2010; Viles 2011; Stallins 2006). The geomorphology of the landforms

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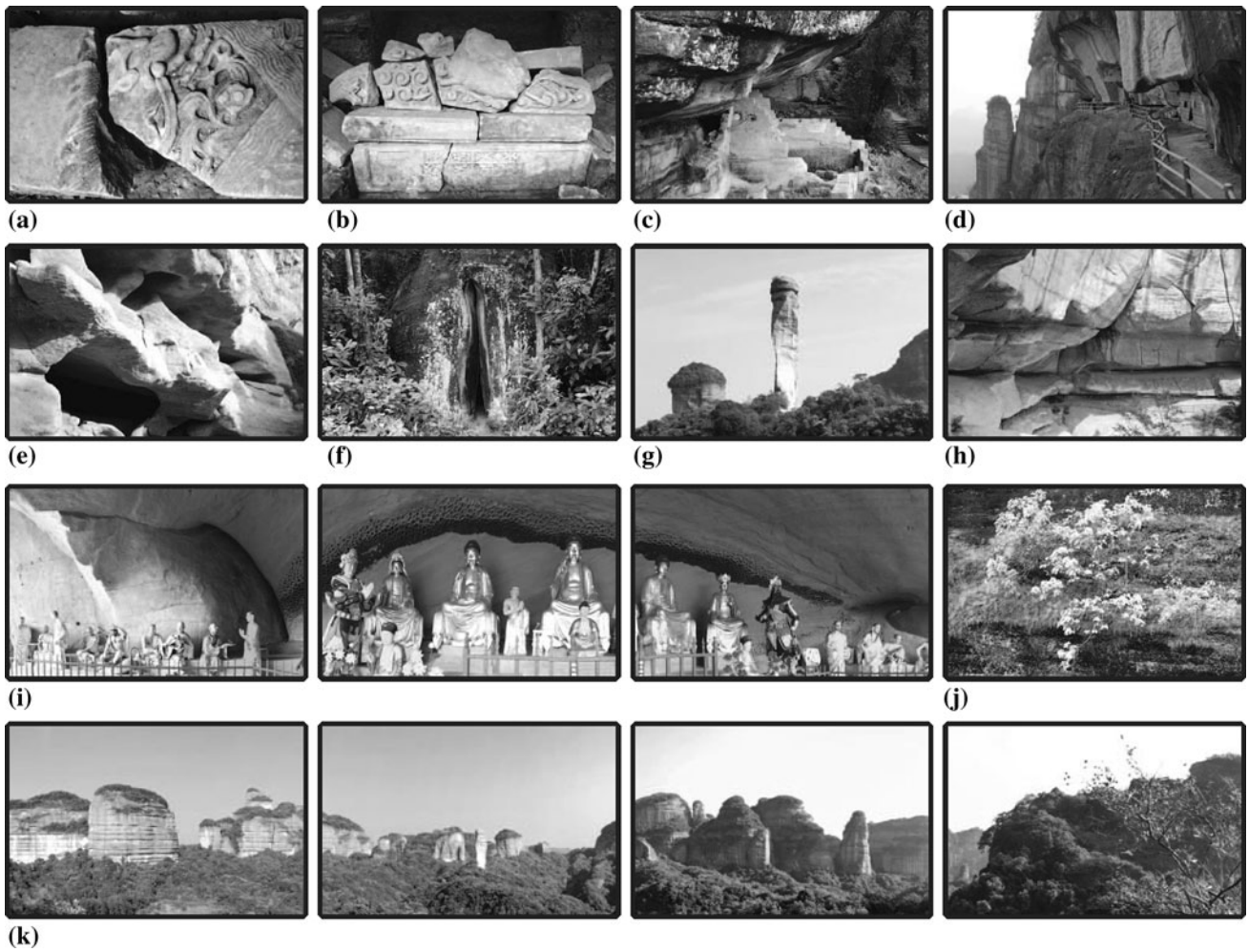

Fig. 1 Danxiashan: **a** Stone Coffin Cave, **b** the Lion Cave altar, **c** Lion Cave **d** the Robber Baron's Hideout, **e** the Jade Screen, **f** Yinyuanshi, **g** Candlestick Peak, **h** Bazhai Gorge, **i** the Color Changing Dragon in the Jinshiyan Nunnery, **j** Firmiana danxiaensis, **k** the view from Wolong Gang

has shaped the landscape's unique ecosystems. The Danxia landforms, formed from uplifted and subsequently eroded red beds, support southern subtropical monsoon forests that are far beyond their climatic maximum (Peng et al. 2008; Hou et al. 2008). Its ecosystems are not only a product of geomorphic patterns and processes, but also play a role in regulating its geomorphology through pedogenesis, biogeochemical cycling, erosion control, slope stabilization, and hydromorphological controls. The climax communities of the Danxia piedmont—evergreen, broad-leaved, southern subtropical monsoon forests rich in tropical ferns and lianas—are dependent upon the microclimate created by the landscape's peculiar geomorphology. North of their climatic maximum, they are a geo-climax community (Peng et al. 2008; Hou et al. 2008). The temperate, broad-leaved, evergreen forests atop the rounded peaks tower above the southern subtropical monsoon forests in the ravines and foot gulches of the Danxia piedmont. Separated by steep cliff faces, the communities of the peaks and of the piedmont below vary in composition and are marked by a radical difference in microclimate. Acting, due to their geomorphology, as isolated habitat islands within a subtropical matrix, the peaks and ridges of Danxiashan are an exemplar of terrestrial island biogeography (Wu et al. 2008). The bio-geomorphic conditions support endemic species such as the Danxia Wutong—Firmiana danxiaensis—that clings to the sheer cliffs of the red-bed peak forests (Fig. 2).

In addition to its geomorphological. ecological, biogeomorphological heritage, Danxiashan has a rich sociocultural legacy. It is culturally significant for its history as a center of Taoism, as pilgrimage destination, and for its mythological characters. Sites of architectural importance such as ancient monasteries, cave temples, hermitages, and old pilgrimage sites dots its cliffs, peak tops, and valleys. The landscape is famous for its symbolic and mystical

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Fig. 2 Danxiashan: terrestrial island biogeography

aesthetics and painterly views—the lush green peaks contrast strongly with the reds of the steep cliffs and the blues of the lakes and rivers that meander through the valleys and ravines along the NNE–EW fault lines (Ministry of Housing and Urban-Rural Development of the People's Republic of China 2008).

In Danxiashan the valleys are laced with fields and are home to villages; the peaks are riddled with cave temples and crowned with great monasteries; and landforms have been given geocultural place names, have been imbued with great cultural significance. Geomorphic, ecological, and socio-cultural processes are linked in adaptive cycles over a range of spatiotemporal scales. Therefore, conservation planning for Danxiashan should not decouple anthropogenic and bio-geomorphic factors.

Even though they may be quantified from a wide range of anthropic values, geomorphosites are focused on geomorphology and applications like geoconservation and geotourism; they are strongly biased toward geophysical scientific values and geocultural heritage. However, in a context like Danxiashan, the ecological, geomorphological, and sociocultural systems are fundamentally and inextricably interconnected. To consider any one—be it ecology, geomorphology, or culture—in isolation would decontextualize it, devalue it, and obscure important processes. If these interconnected systems are assessed on an equal footing, then they can be rationally and systematically weighted with multi-criteria decision analysis according to context-specific strategic management objectives, rather than being biased by disciplinary prejudices. Thus, a bespoke assessment strategy for Danxiashan requires that the concept of geomorphosites be adapted to value ecological, geomorphological, and cultural criteria equally so that geomorphosites become biogeocultural sites.

Any collection of geomorphosites or heritage sites can be considered a network. A network may be purely conceptual, a collection of landforms linked by a common classification, or it may be physically linked and spatially interconnected like the branches of a river system or highways in a road network. Geomorphosites and networks of geomorphosites occur across spatial scales and may be nested. For example, Danxiashan is part of a global network of World Heritage sites (Ministry of Housing and Urban-Rural Development of the People's Republic of China 2008), it is part of a regional network of Danxia geomorphosites (Jin 2011), and it contains a network of heritage sites ranging from hoodoo columns to Taoist cave temples. While the concept of geomorphosites can be used as is to identify opportunities for geotourism, spatial connections between geomorphosites should be considered in geotourism prioritization and planning.

4 Methodology

4.1 Geospatial planning

The first step in our planning process is to define the decision-makers' and stakeholders' objectives. An objective translates abstract, theoretical values into an applied

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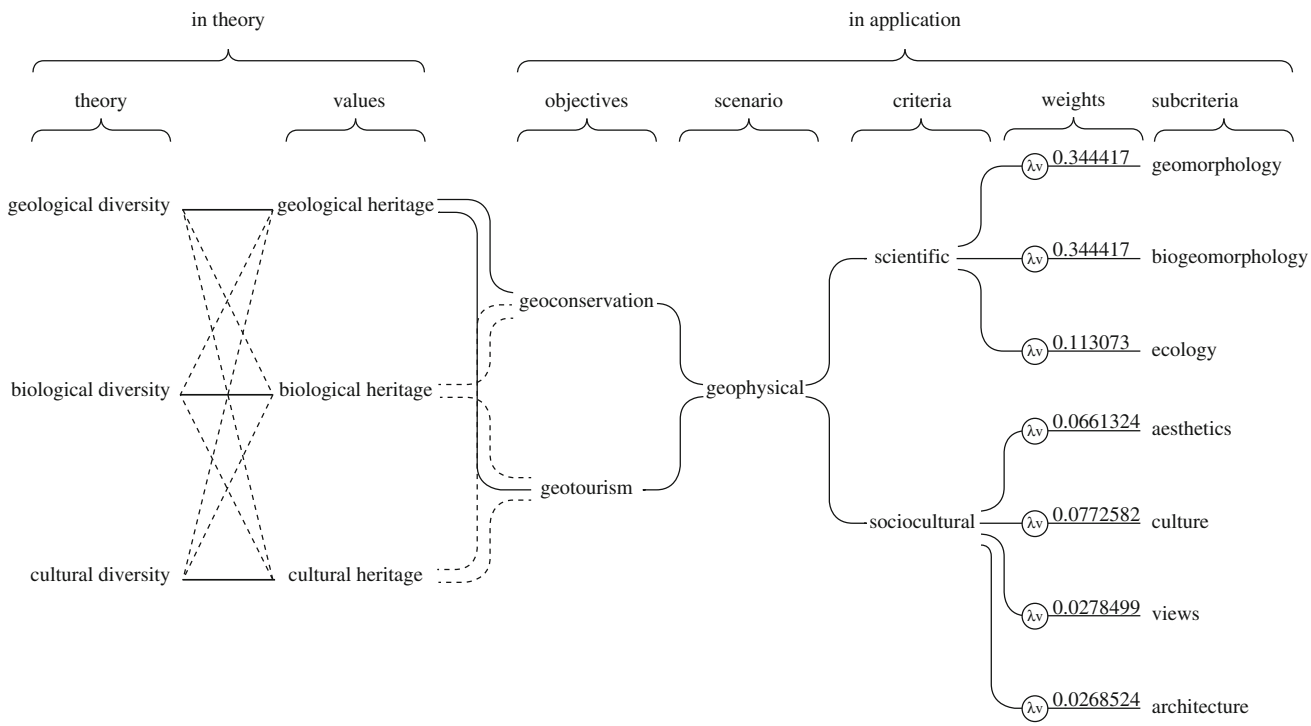

Fig. 3 Conceptual model for Scenario 1: geophysical preference

design problem. After defining potential objectives we offer scenarios that seek to balance a set of objectives. Each scenario is then categorized by criteria and weighted sub-criteria that represent the theoretical values behind the objectives. All of the sites are then assessed and evaluated for each sub-criteria using quantitative metrics. Once assessed the sites can be prioritized for each scenario through multi-criteria decision analysis. Finally an optimal network is computed for each scenario using least-cost path and networking analyses.

The objectives of our study represented a set of abstract values—geological heritage, biological heritage, and cultural heritage—derived from a set of theoretical concepts—geological diversity, biological diversity, and cultural diversity. Our objectives—geological conservation, geological tourism, biological conservation, biological tourism, heritage conservation, and heritage tourism—are applications of these abstract values. Scenario 1, the geophysical scenario, balanced geoconservation and geotourism objectives (Fig. 3). Scenario 2, the socio-cultural scenario, balanced heritage conservation and tourism. Scenario 3, the biogeocultural scenario, balanced all of the objectives.

4.2 Spatial decision support

We used a prototype spatial decision support system (SDSS) to design a trail network linking select heritage sites in Danxiashan. When designing a trail network oft-conflicting objectives will be considered in geographic space; it is no longer just a question of rationally comparing criteria, it is also a spatial problem requiring combinatorial optimization. SDSS provide a computational framework for decision-making about geospatial issues. They are designed to help managers solve complex spatial problems through analysis, planning, scenarios, and prioritization (University of Redlands and Spatial Decision Support Consortium 2012). A GIS-based SDSS using multi-criteria decision analysis was used to weight decision criteria, prioritize heritage sites, optimize the trail networks, and generate scenarios for geotourism in Danxiashan.

The GIS data used in this research was derived from the ASTER Global Digital Elevation Model (METI and NASA 2011), management plans (Planning and Design Institute of Sun Yat-sen University 2009), tourist statistics (Danxiashan Administrative Committee 2011), and heritage site assessments conducted by one of the authors (B. Harmon).

4.3 Heritage site assessment

4.3.1 Metrics

While there are many potential heritage sites in Danxiashan, due to the limited scope of this study we selected 37 sites that were already part of the existing system of tourist trails in the core tourism area. These sites were assessed by one of the authors with input by experts—including ecologists and geomorphologists—and were then evaluated using an array of metrics, ordered in a hierarchical structure. Categories

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included scientific criteria and socio-cultural criteria. The scientific sub-criteria were biogeomorphology, ecology, and geomorphology. The sociocultural sub-criteria evaluated were aesthetics, views, architecture, and cultural heritage. The architectural sub-criterion was further defined by the following sub-sub-criteria: climate, context, cultural heritage, light, tectonics, and topography. Each criterion was evaluated on a scale from 1 to 4 that represented a degree of rarity defined by characteristics particular to the value. While we used a numeric scale, we believe that it would be better to use a fuzzy scale ranging from true to false, described by semantic qualifiers.

The integrity of each of the socio-cultural and scientific sub-criteria was also assessed. UNESCO defines integrity as "a measure of the wholeness and intactness of the natural and/or cultural heritage and its attributes" (UNESCO 2011). We used metrics of preservation—a measure of intactness—and sensitivity—a measure of susceptibility to risk—to quantify integrity. For example the heritage values of a site may have been undermined, its integrity compromised, if it has been damaged by overexploitation, over-trafficking, graffiti, littering, erosion, weathering, etc. Furthermore a site that is currently intact but is being visited by an ever increasing number of visitors may be at high risk of deterioration (See the electronic supplementary material for ranking metrics).

4.3.2 Weighted scenario rankings

Different geotourism objectives—representing potential stakeholder viewpoints—were explored in three scenarios. In the first scenario scientific, geophysical values were preferred; in the second, socio-cultural values; and in the third, biogeocultural interactions. Scenario scores for each site were determined by the sum of their heritage values weighted by preference and integrity. First, each of the heritage criteria was weighted according to its preservation and sensitivity scores. Then for each scenario the analytic hierarchy process was used to weight criteria by preference through pairwise comparison. Once the criteria had been weighted by their eigenvectors, they were summed to create a scenario score for each site using Eq. (1).

$$S_x = \sum_{y=1}^{7} c_{y_x} \frac{p_{y_x} + s_{y_x}}{8} \lambda v_{y_x} \tag{1}$$

S scenario score, c criterion score, p preservation score, s sensitivity score, λv eigenvector.

4.4 Least-cost path analysis

4.4.1 Landscape suitability

Planning a trail network through such a complex landscape requires that often-conflicting objectives or criteria are compared and weighed. Fuzzy logic in map overlay analysis was used to create a suitability map for the landscape, a map of how suitable the terrain is, cell by cell, for potential trails (Fig. 4). In fuzzy overlay analysis raster maps are evaluated with fuzzy logic operations and then merged to create a composite raster map, a suitability map. Fuzzy logic is used to address issues about imprecision and vagueness in the classification of data in spatial multi-criteria decision analysis (Esri 2012a; Raines et al. 2010).

Suitability was determined by merging a complex array of anthropic and environmental criteria through fuzzy logic operations. These criteria include: (a) minimizing travel distance and construction costs by preferring low slopes, (b) maximizing opportunities for prime views, (c) conserving biogeographic islands, (d) protecting rare and endemic species, (e) protecting rare and endemic habitats, (f) conserving ecological communities, and (g) minimizing anthropogenic impacts. The geospatial layers used in this map overlay analysis were derived from the ASTER Global Digital Elevation Model and from surveys, land classification, and land use mapping by Sun Yat-sen University. The biogeography layer represents peak top communities. The rare and endemic species layers represent sites where threatened species have been recorded while the rare and endemic habitat layers represent plant communities that include threatened species. The anthropogenic impacts layer, derived from land use, shows where infrastructure and development have degraded the ecological integrity of the landscape. As fuzzy criteria these layers were each classified on a gradient from true to false.

4.4.2 Iterative least-cost path algorithm

An iterative least-cost path algorithm was used to generate potential routes to heritage sites in Danxiashan. Cost distance functions were used to assign each cell of the study area an accumulative cost based on the travel distance over the suitability map from the last source, a heritage site, to the next destination, another site. In each iteration, a least-cost path function was used to find the optimal route over an accumulated cost surface from a designated source to a designated destination; with each run the previous destination became the new source, thus sequentially linking every heritage site (Esri 2011). Expert judgment was used to further refine the trail system, adding additional circuitry to increase connectivity.

4.5 Networking scenarios

Alternative trail systems were designed by combinatorial optimization. For each networking scenario, heritage sites were selected if they had a median scenario score or better. Route analysis (Esri 2012b) was then used to find the optimal route between select heritage sites over the

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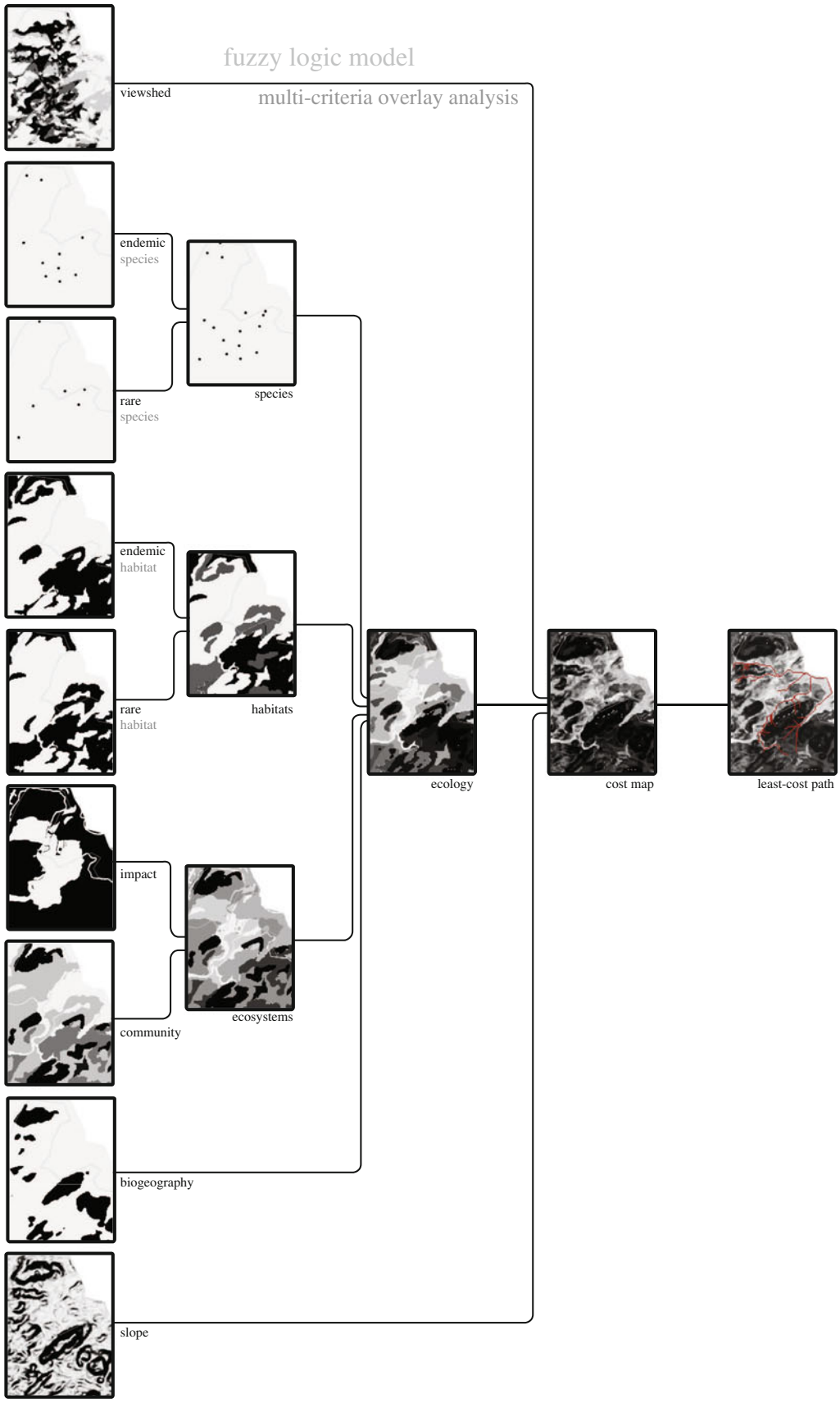

Fig. 4 Fuzzy logic model for map overlay analysis (data sources: METI & NASA and Sun Yat-sen University)

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network of potential trails generated by the iterative least-cost path algorithm. When presented with scenarios for alternative trail systems based on their preferences, decision-makers can compare results, re-analyze their preferences and make trade-offs, compare new iterations, and potentially reach group consensus.

Scenarios also can be used to explore alternative designs. For example if stakeholders are interested in creating a didactic experience—focused on geo-education, traditional culture, and bio-geomorphic interactions—they may consider planning multiple thematic routes. In this scenario (Fig. 5) three thematic routes have been generated: (a) a geomorphic route, (b) a cultural route, and (c) a heritage route. Routes were generated by solving a location-allocation problem. For each scenario, sites with median scenario scores or above were selected as candidates and weighted by their score. Sites with median or above scores in other scenarios were selected as competitors and weighted. Sites were then chosen for each route by calculating the minimum selection of candidate sites required to obtain at least 15 % of the market. Routes were optimized for travel-time by solving a vehicle routing problem for visitors who had 8 h to explore Danxiashan and spent 15 min per site.

With separate thematic routes visitors would have the opportunity to learn about themes like geoheritage in more depth because each route could have more specialized and topical signage and information. Furthermore, if the spatial decision support system was implemented as a web-based smart phone app then tourists could use it to plan custom, personalized itineraries. A similar system for planning customized tourist itineraries was developed for visitor centers in the Great Smoky Mountains National Park (Dye and Shaw 2007).

5 Results

A series of scenarios—three scenarios representing different stakeholder values (Fig. 6) and a scenario with thematic routes (Fig. 7)—were explored through the prototype SDSS. By first quantifying the multiplicity of heritage values in each site and then weighting the values with the analytic hierarchy process, we were able to prioritize conservation and tourism for specific objectives representing potential stakeholder views, that is, a geophysical preference (2), a cultural and aesthetic preference (3), and biogeocultural preference (4). These preferences were derived from the eigenvectors of near consistent matrices of judgements based on pairwise comparison. To practically conserve our heritage and our environment we must balance conservation and development pressures; when there are conflicting objectives, trade-offs must be made. While tourism can conflict with conservation, it can also, synergistically, promote education, engagement, and investment that can further long-term conservation goals. These scenarios are an example of how tourism and conservation can be synergistically balanced.

Scenario 1: Geophyscial

$$\begin{bmatrix} & c_1 & c_2 & c_3 & c_4 & c_5 & c_6 & c_7 \\ c_1 & 1 & 3 & \frac{1}{6} & 1 & \frac{1}{2} & \frac{1}{6} & 3 \\ c_2 & \frac{1}{3} & 1 & \frac{1}{9} & \frac{1}{6} & \frac{1}{4} & \frac{1}{9} & 1 \\ c_3 & 6 & 9 & 1 & 6 & 4 & 1 & 9 \\ c_4 & 1 & 3 & \frac{1}{6} & 1 & \frac{1}{2} & \frac{1}{6} & 3 \\ c_5 & 2 & 5 & \frac{1}{4} & 2 & 1 & \frac{1}{4} & 5 \\ c_6 & 6 & 9 & 1 & 6 & 4 & 1 & 9 \\ c_7 & \frac{1}{3} & 1 & \frac{1}{9} & \frac{1}{3} & \frac{1}{5} & \frac{1}{9} & 1 \end{bmatrix} \tag{2}$$

c_1 aesthetics score, c_2 architecture score, c_3 biogeomorphology score, c_4 culture score, c_5 ecology score, c_6 geomorphology, c_7 views.

Results: Scenario 1

Maximum eigen value = 7.240

Consistency index = 0.040

Weights

$c_1 = 0.066$, $c_2 = 0.0269$, $c_3 = 0.344$

$c_4 = 0.077$, $c_5 = 0.113$, $c_6 = 0.344$, & $c_7 = 0.028$

Scenario 2: Cultural & Aesthetic

$$\begin{bmatrix} & c_1 & c_2 & c_3 & c_4 & c_5 & c_6 & c_7 \\ c_1 & 1 & \frac{1}{3} & 4 & \frac{1}{3} & 7 & 5 & 2 \\ c_2 & 3 & 1 & 6 & 1 & 9 & 7 & 4 \\ c_3 & \frac{1}{4} & \frac{1}{6} & 1 & \frac{1}{6} & 3 & 2 & \frac{1}{3} \\ c_4 & 3 & 1 & 6 & 1 & 9 & 7 & 4 \\ c_5 & \frac{1}{7} & \frac{1}{9} & \frac{1}{3} & \frac{1}{9} & 1 & \frac{1}{3} & \frac{1}{6} \\ c_6 & \frac{1}{5} & \frac{1}{7} & \frac{1}{2} & \frac{1}{7} & 3 & 1 & \frac{1}{4} \\ c_7 & \frac{1}{2} & \frac{1}{4} & 3 & \frac{1}{4} & 6 & 4 & 1 \end{bmatrix} \tag{3}$$

c_1 aesthetics score, c_2 architecture score, c_3 biogeomorphology score, c_4 culture score, c_5 ecology score, c_6 geomorphology, c_7 views

Results: Scenario 2

Maximum eigen value = 7.294

Consistency index = 0.049

Weights

$c_1 = 0.154$, $c_2 = 0.314$, $c_3 = 0.051$, $c_4 = 0.314$,

$c_5 = 0.022$, $c_6 = 0.038$, & $c_7 = 0.106761$

Scenario 3: Biogeocultural values

$$\begin{bmatrix} & c_1 & c_2 & c_3 & c_4 & c_5 & c_6 & c_7 \\ c_1 & 1 & 2 & \frac{1}{4} & 2 & \frac{1}{3} & \frac{1}{3} & 2 \\ c_2 & \frac{1}{2} & 1 & \frac{1}{5} & 1 & \frac{1}{4} & \frac{1}{4} & 1 \\ c_3 & 4 & 5 & 1 & 5 & 2 & 2 & 5 \\ c_4 & \frac{1}{2} & 1 & \frac{1}{5} & 1 & \frac{1}{4} & \frac{1}{4} & 1 \\ c_5 & 3 & 4 & \frac{1}{2} & 4 & 1 & 1 & 4 \\ c_6 & 3 & 4 & \frac{1}{2} & 4 & 1 & 1 & 4 \\ c_7 & \frac{1}{2} & 1 & \frac{1}{5} & 1 & \frac{1}{4} & \frac{1}{4} & 1 \end{bmatrix} \tag{4}$$

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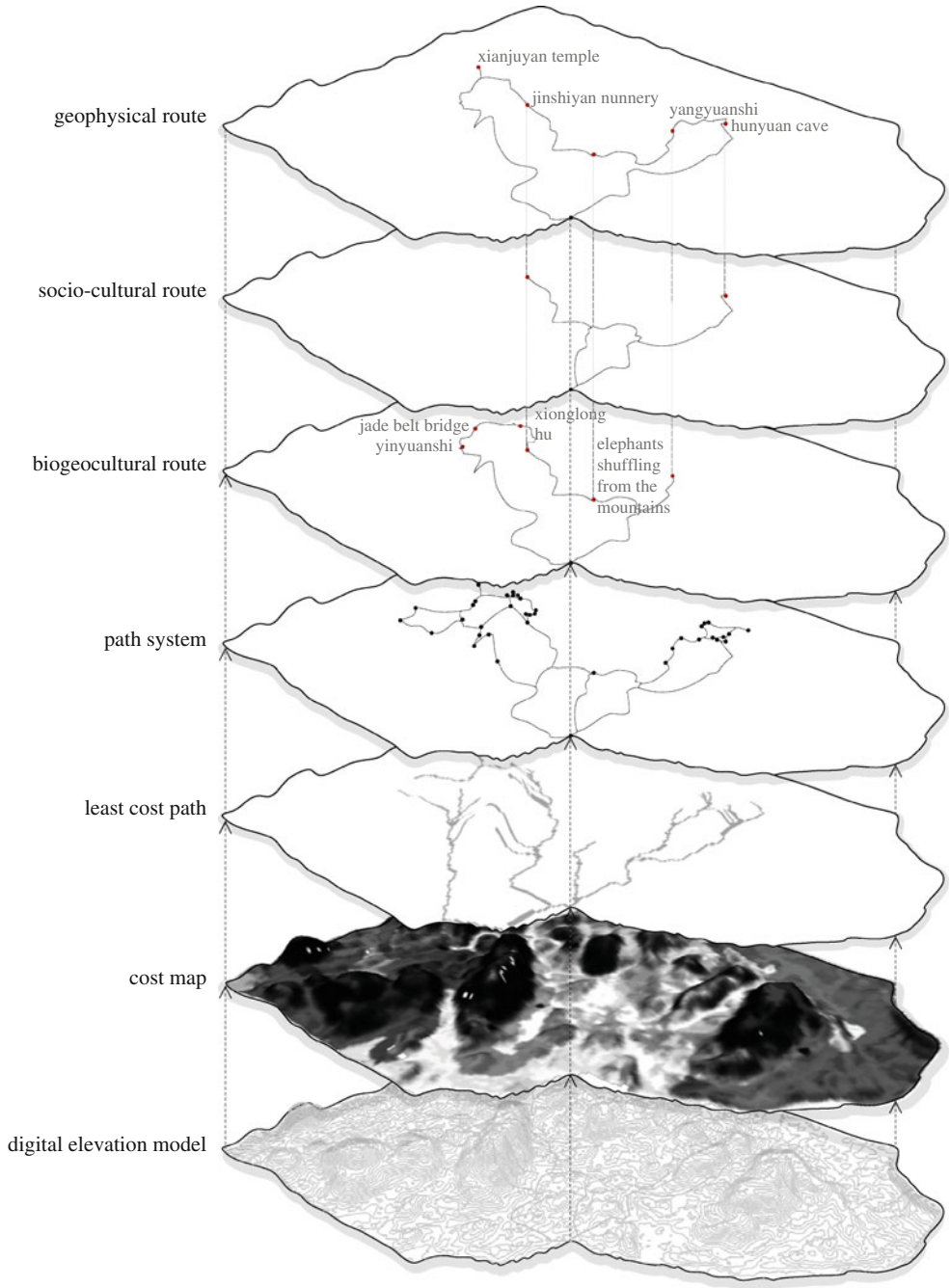

Fig. 5 Networking workflow (data sources: METI & NASA and Sun Yat-sen University)

c_1 aesthetics score, c_2 architecture score, c_3 biogeomorphology score, c_4 culture score, c_5 ecology score, c_6 geomorphology, c_7 views

Results: Scenario 3

Maximum eigen value = 7.068

Consistency index = 0.011

Weights

$c_1 = 0.090$, $c_2 = 0.053$, $c_3 = 0.053$, $c_4 = 0.329$, $c_5 = 0.211$, $c_6 = 0.053$, & $c_7 = 0.053$

6 Discussion

By weighting criteria based on their objectives and preferences, decision-makers can participate in the modeling and design process. By comparing alternative trail systems, decision-makers can test their preconceptions, reanalyze their objectives, make trade-offs, re-weight their preferences, and, ideally, build group consensus, based on science, through collaboration. If diverse groups of stakeholders were included in these scenario workshops, this

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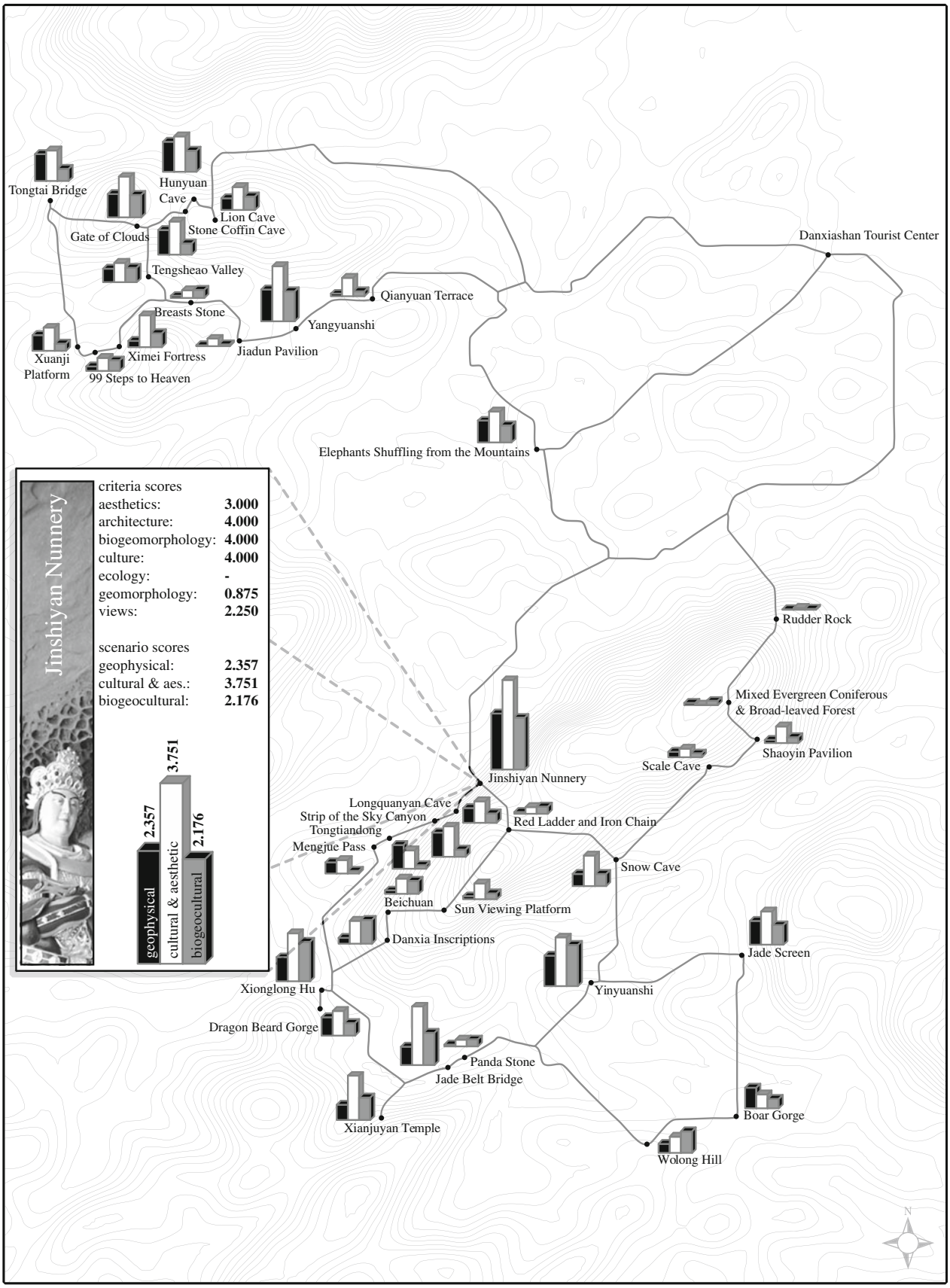

Fig. 6 Scenario 3: Biogeocultural (data sources: METI & NASA and Sun Yat-sen University)

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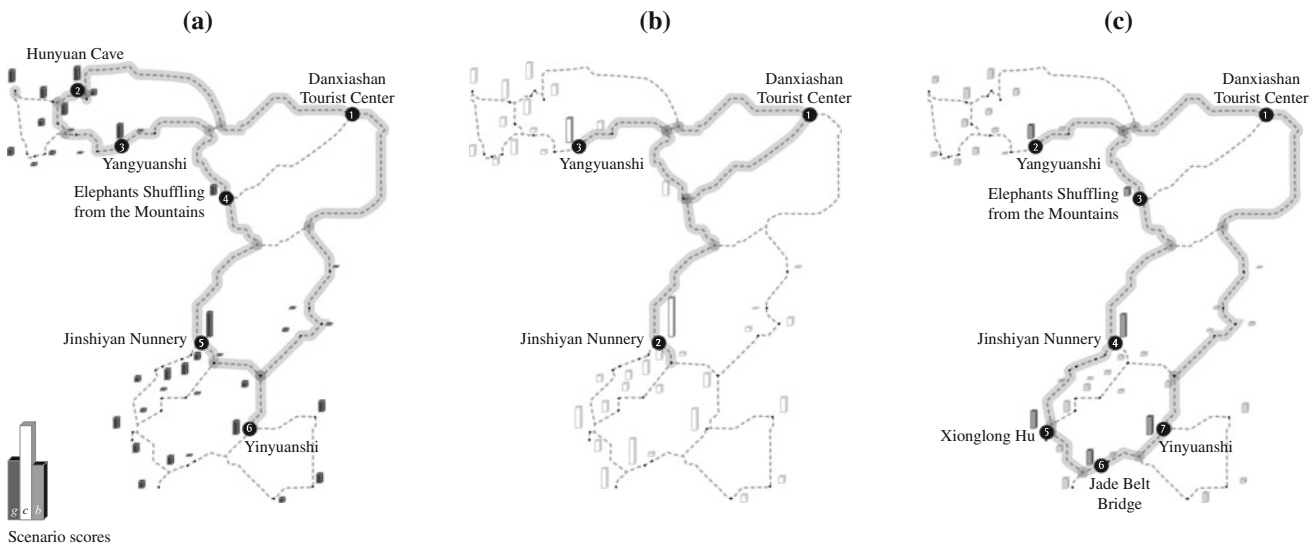

Fig. 7 Thematic route scenario: **a** geophysical, **b** sociocultural, & **c** biogeocultural routes (data sources: METI & NASA and Sun Yat-sen University)

iterative cycle of participatory modeling and scenario analysis could lead to a more open, transparent, and democratic design and planning process. Engaging stakeholders in the planning process may build support for the project, nurture conservation ethics, and promote geo-education. Participatory modeling can also encourage cross-sector collaboration and exchange, promoting strategic thought across disciplines (Sandker et al. 2010).

This method for designing networks of heritage sites requires high quality, high-resolution data. Inaccurate or low-resolution data will distort the results. Since slope is one the primary constraints in trail design, high-resolution topographic data is needed to accurately design trails in landscapes with complex landforms and near-vertical slopes like Danxiashan. For example, in digital elevation models derived from orbital remote sensing a cluster of small Danxia peaks may be interpolated into a single landform. Ideally, high-resolution topographic data for such a study would be collected by airborne LIDAR missions and then verified by ground-truthing.

Data quality and resolution are not the only sources of significant inaccuracies; inaccuracies in data can arise through classification. Boundaries between classes may be vague, ill defined, or imprecise; classifying data into discrete classes may obscure important patterns and processes (Raines et al. 2010). For example, ecosystems often do not have crisp boundaries—instead they may have an ecological gradient or ecotone between them. If the ecosystem boundary is interpolated through the middle of the ecotone, a critical ecological process will be obscured. Such imprecise, subjectively defined, semantically described boundaries can be objectively quantified using fuzzy logic.

To address inaccuracies arising from the classification of data, fuzzy logic could be used throughout the geomorphosite or heritage site assessment and planning process. Heritage metrics could be assessed with semantic qualifiers by multiple experts and classified based on their degree of membership, spatial data could be classified and merged in fuzzy map overlay analysis, and criteria could be weighted with the fuzzy analytic hierarchy process. With fuzzy logic, qualitative feedback from stakeholders could also be integrated, enabling better public engagement.

While a network of heritage sites may be designed to conserve heritage, minimize ecological and geomorphological impacts, and maximize opportunities for tourism and education at the time of its planning, its context will continue to evolve. In Danxiashan the landforms and ecosystems will continue to coevolve: as peaks erode, microclimates will change and reshape community assemblages; entire habitat islands will be lost in mass-wasting events as cliffs give way. Furthermore, tourist will compact soils, accelerate erosion, and degrade sensitive sites. Management will be required to conserve the heritage sites. While heritage site planning can provide a initial, sound basis for management in Danxiashan, recent black swan events have shown that modeling alone is not a sufficient guide for long-term management. Black swan events are rare, seemingly unpredictable events that have a disproportionately large impact and are rationalized in hindsight. As statistical outliers, black swan events are unlikely to be predicted through modeling although methods like reference class forecasting may reduce risk (Flyvbjerg and Budzier 2011). By seeking to build resilience rather than focusing solely on avoiding risks, rare, high-impact events including natural disturbances can be

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weathered, can be adapted to. Natural disturbance regimes and large infrequent disturbances like hurricanes are a crucial part of long-term ecosystem dynamics (Bengtsson et al. 2003).

Given the complexity, stochasticity, and unpredictability of ecosystems subject to hierarchical disturbances, diverse disturbance legacies, and multiple post-disturbance pathways, management should build resilience, conserve ecological memory, accept the need for natural disequilibrial change, and seek to mitigate anthropogenic degradation (Mori 2011). Adaptive management—an iterative process of monitoring, learning from feedback, and tailoring management actions to suit changing conditions (Salafsky and Margoluis 2004)—reduces uncertainty, mitigates risk, and optimizes decision-making (Nichols et al. 2007). While traditional, environmental management seeks to maintain equilibrium, adaptive management, by adapting to and learning from change, can regulate a system in dynamic equilibrium or in disequilibrium, encouraging beneficial successional pathways and ecosystem change while mitigating harmful anthropogenic stressors (Mori 2011).

While adaptive management can be used to regulate social, geomorphic, and ecological processes, potential interventions are constrained by the static design of the park. The trail system at Danxiashan could be optimized initially through heritage site planning and networking and then both the sites and the greater landscape could be conserved, dynamically, with adaptive management. However, once a site or a patch of the landscape changes then the trail system may no longer be optimal. Any piecemeal changes to spatial catalysts or design criteria (Pressey et al. 2007) for the sake of conserving ecological or geomorphological patterns will not be systematic unless the entire structure of the park is reconsidered.

7 Conclusions

To efficiently and effectively conserve our natural heritage, conservation efforts must be systematically planned and prioritized. By adapting the concept of geomorphosites to represent the multivalent human values for a landscape, conservation challenges and the potential for tourism and education can be better recognized, evaluated, and prioritized. A heritage planning and networking process based on multi-criteria site assessment and decision analysis, combinatorial optimization, and scenario planning can be used to prioritize heritage conservation and geotourism in complex landscapes with a multiplicity of values.

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